

input

state

$$\mathbf{S}_t \in \mathbb{Z}_{\geq 0}^d$$

normalization

$$\mathbf{x} = \mathbf{S}_t / \kappa$$

backbone $u_\theta(\mathbf{x})$

linear | NN_H | soft tree

decoder

(gated / ordinal /
categorical / tree)

output

valid action

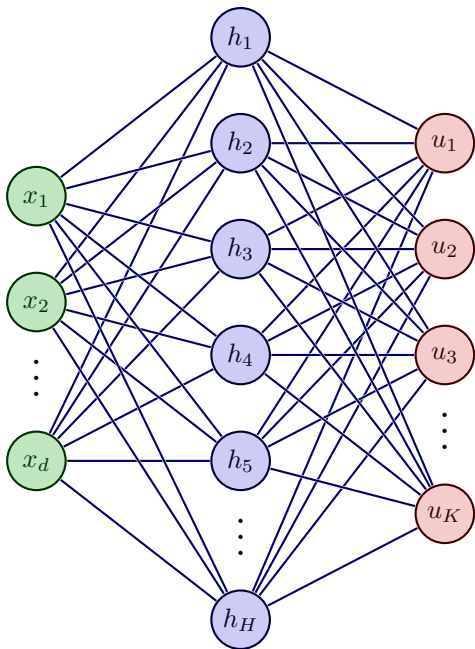
$$a_t$$

policy π_θ (parameters θ learned by CMA-ES)

normalized
state $\mathbf{x}$

hidden layer
(H units)

latent
outputs $\mathbf{u}$



latent
outputs
 $\mathbf{u}$

categorical (scores $y_0, \dots, y_{\bar{q}}$):

$$a = \arg \max_{q \in \{0, \dots, \bar{q}\}} y_q$$

soft-gated direct (gate g , quantity q):

$$a = \lfloor \sigma(g) \cdot \text{softplus}(q) \rfloor_{\geq 0}$$

soft-gated ordinal (gate g , thresholds $o_1, \dots, o_M$):

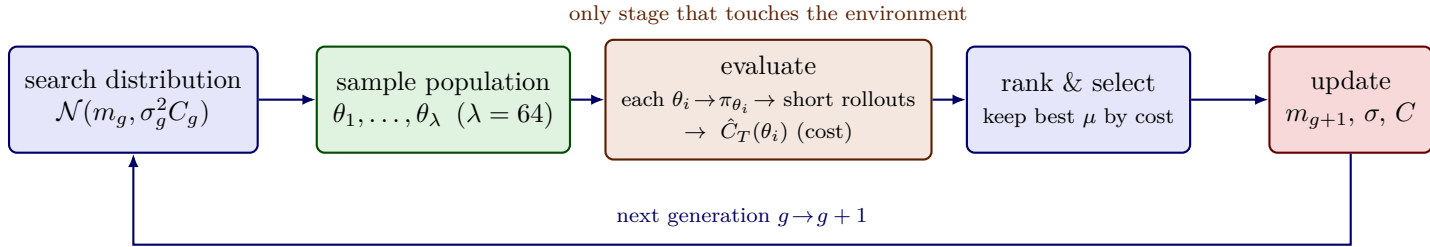
$$a = \left\lfloor \sigma(g) \sum_{k=1}^M \sigma(o_k) \right\rfloor_{\geq 0}$$

soft tree (leaf prob. π_ℓ , leaf quantity q_ℓ):

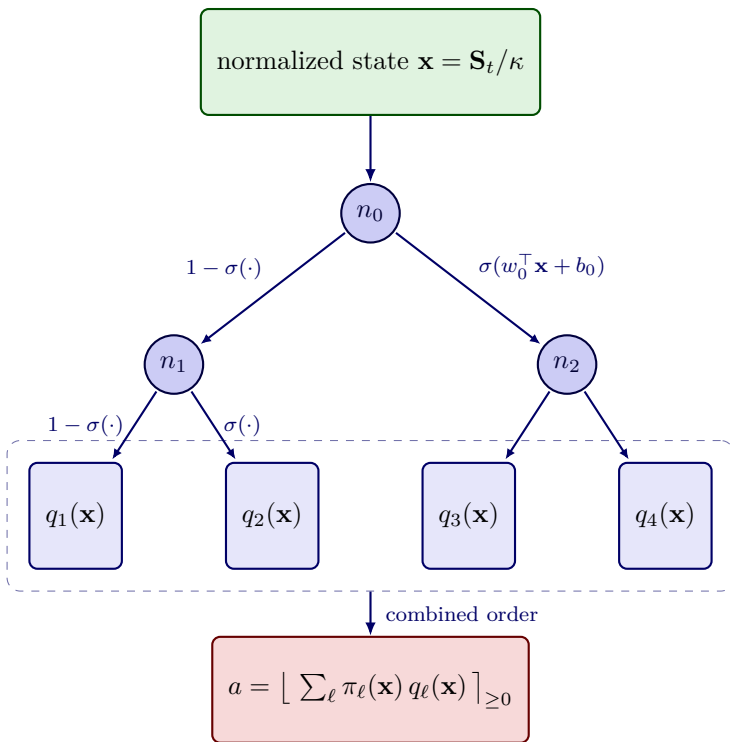
$$a = \left\lfloor \sum_{\ell} \pi_{\ell}(\mathbf{x}) q_{\ell}(\mathbf{x}) \right\rfloor_{\geq 0}$$

**valid
action a_t**

σ : logistic; $\text{softplus}(z) = \ln(1 + e^z)$; $\lfloor \cdot \rfloor_{\geq 0}$: round to nearest non-negative integer.

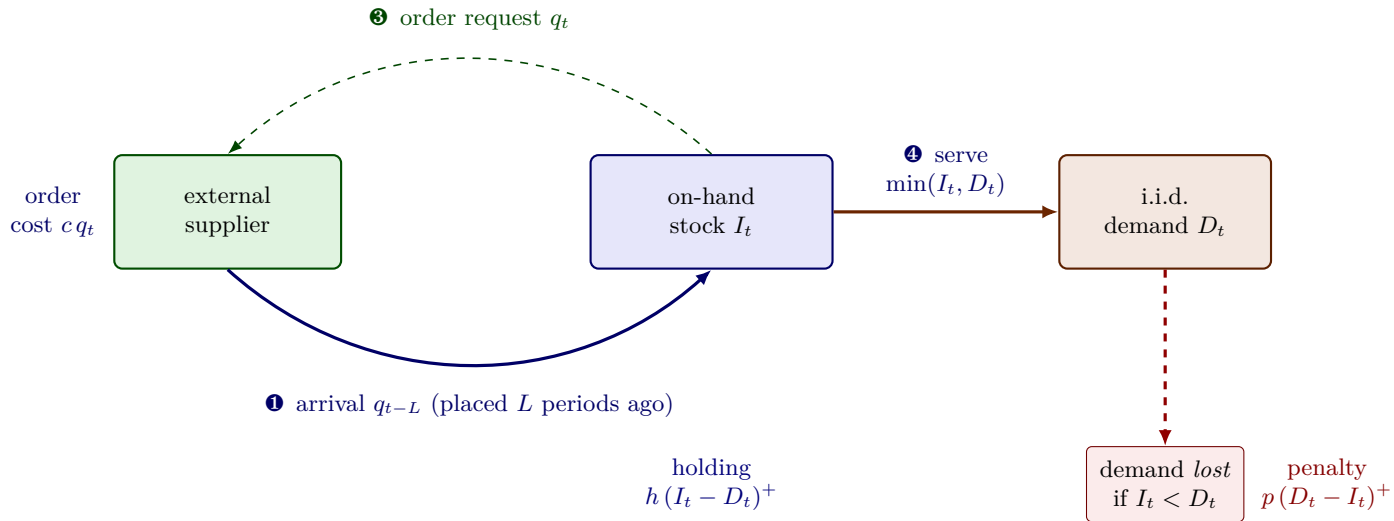


Gradient-free: the update uses only the *ranking* of the costs $\hat{C}_T(\theta_i)$, not their gradients.



Each leaf ℓ has path probability $\pi_{\ell}(\mathbf{x}) = \prod(\text{routing probs on its root-to-leaf path})$, with $\sum_{\ell} \pi_{\ell} = 1$; the action is the soft mixture of the leaf quantities, rounded to a valid order.

Lost-sales environment (one period)



Per-period event sequence

➊ arrival q_{t-L} joins on-hand stock

➋ observe post-arrival state $\mathbf{S}_t$

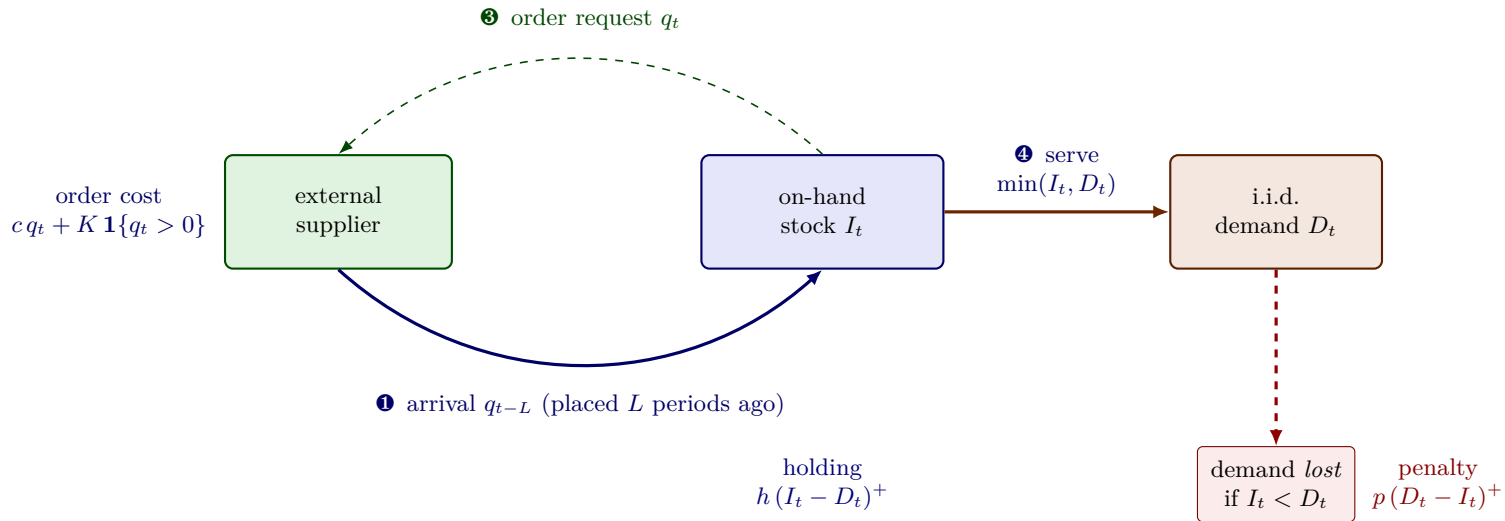
➌ place order request q_t (arrives in L)

➍ demand D_t served from on-hand I_t

➎ unmet demand *lost*

➏ charge holding + lost-sales

Fixed-cost lost-sales environment (one period)



Per-period event sequence

➊ arrival q_{t-L} joins on-hand stock

➋ observe post-arrival state $\mathbf{S}_t$

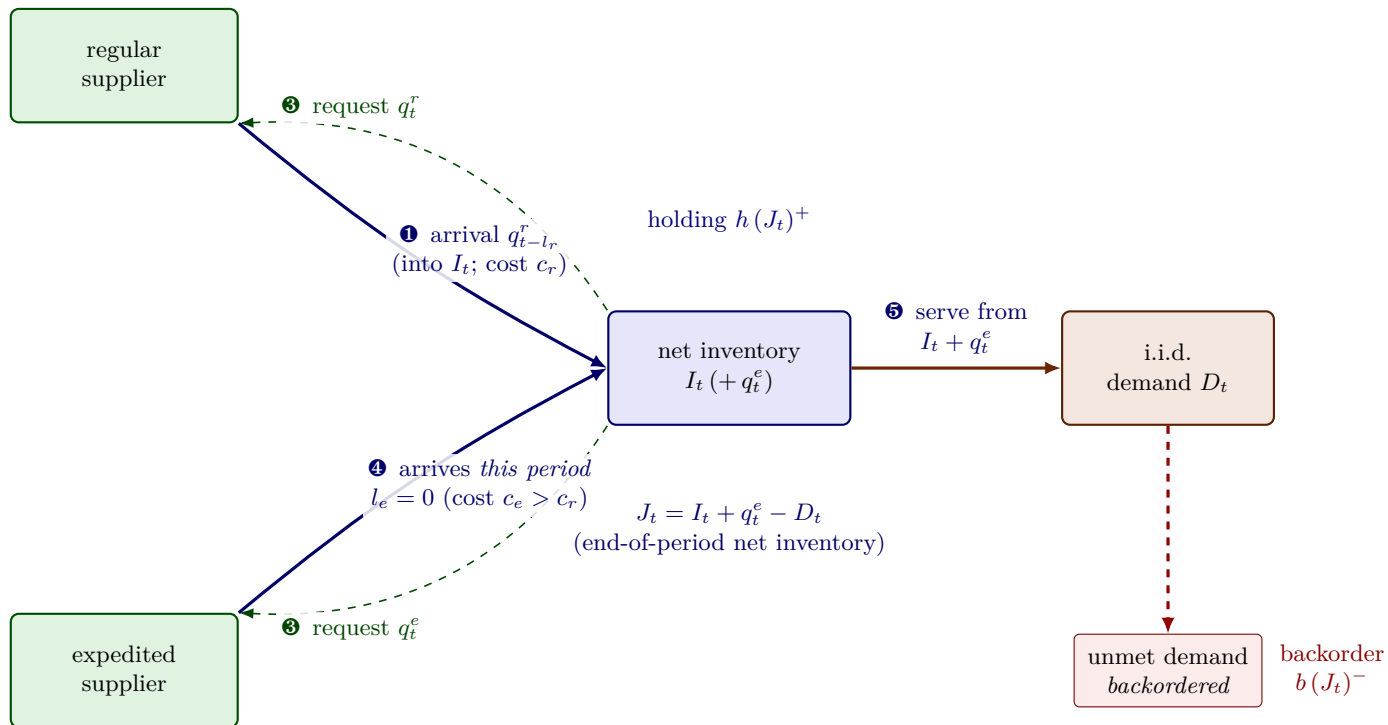
➌ place order q_t ($setup + K$ if $q_t > 0$)

➍ demand D_t served from on-hand I_t

➎ unmet demand *lost*

➏ holding + lost-sales + setup

Dual-sourcing environment (one period)



Per-period event sequence

➊ regular arrival $q_{t-l_r}^r$ joins net inventory I_t

➋ observe state $\mathbf{S}_t$

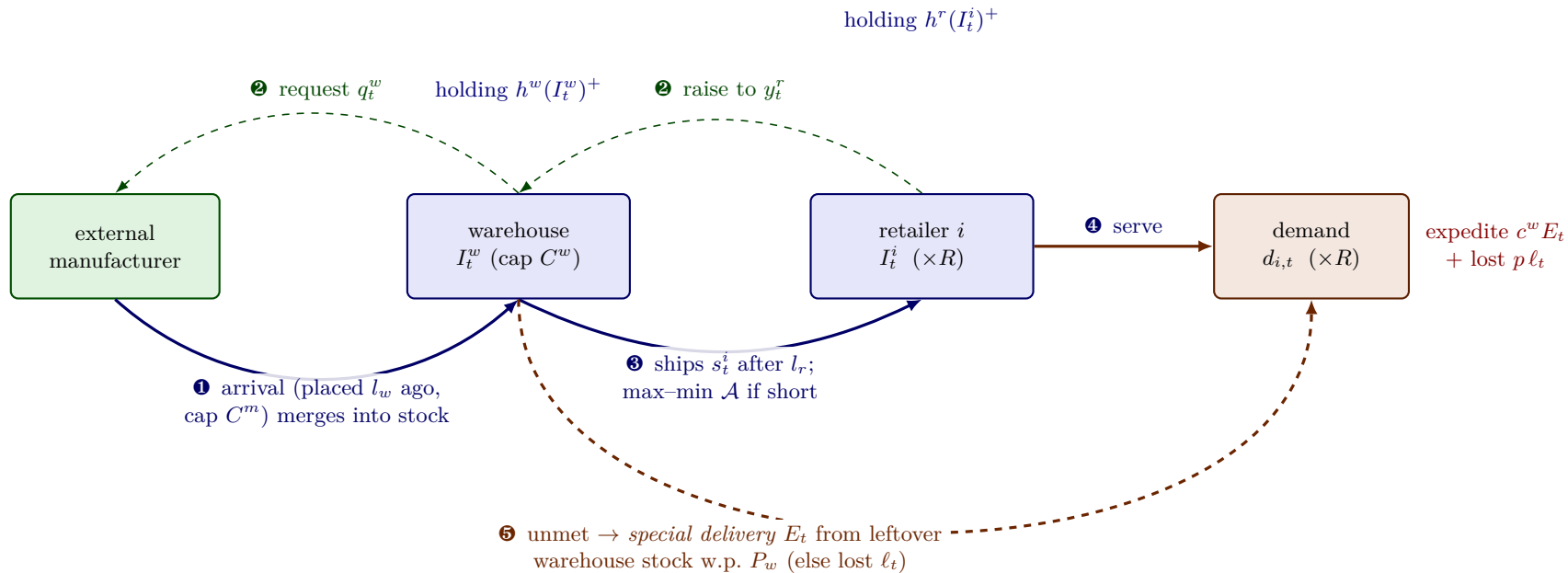
➌ request (q_t^r, q_t^e)

➍ expedited q_t^e arrives *this period*

➎ demand D_t served from $I_t + q_t^e$; unmet *backordered*

➏ procurement + holding + backorder

Divergent multi-echelon environment with special delivery (one period)



Per-period event sequence

➊ arrivals merge into warehouse & retailer on-hand (before ordering)

➋ each retailer serves demand $d_{i,t}$

➌ warehouse orders q_t^w (pre-shipment IP); retailers raise to y_t^r

➍ unmet $\rightarrow$ special delivery from leftover stock (w.p. P_w) else lost

➎ warehouse ships s_t^i (max-min $\mathcal{A}$ if short)

➏ holding + expedite + lost